

The Impacts of HTTPS Adoption For Online Services Located in China

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Introduction

History of HTTPS and China

- ▶ Hypertext Transfer Protocol Secure (HTTPS) combined TLS encryption with HTTP data transfer for secure communication.
- ▶ Domains located in China have historically shown lower-than-average adoption rates for HTTPS and related security metrics.
- ▶ Prior work:
 - ▶ Assessment of the impacts of TLS vulnerabilities in the HTTPS ecosystem in China: collected measurements on the web ecosystem in China and identified various web attacks that can take advantage of TLS vulnerabilities on these domains.¹
 - ▶ Measuring HTTPS Adoption on the Web: explored differences between HTTPS adoption amongst a wide array of categories and found regional disparities to be one of the most significant factors.²
 - ▶ Cloudflare Radar: reports 83% HTTPS utilization rate in China vs. 98% worldwide.³

Objectives for This Study

- ▶ Phase 1: Program applications to automatically request and acquire information related to HTTPS from popular domains across the web.
- ▶ Phase 2: Perform analysis on measured information related to differences in HTTPS adoption between domains located in China vs. other countries.
- ▶ Phase 3: Briefly explore the reasons for and impacts of HTTPS adoption on domains located in China.

Approach

Overview of Tools

- ▶ **Cloudflare Radar:** Initial inspiration and contains statistics on web traffic.
- ▶ **Tranco:** Provides large dataset of most popular web domains.⁴
- ▶ **MaxMind GeoIP:** Database for identifying geographic location of web servers.⁵
- ▶ **Python modules:**
 - ▶ requests - Enables the construction and delivery of HTTPS requests.⁶
 - ▶ socket - Interface for inter-process communication (particularly IPv4).⁷
 - ▶ ssl - Provides a TLS/SSL wrapper for socket objects.⁸
 - ▶ OpenSSL - Cryptographic tool for analyzing TLS connections.⁹
 - ▶ geoip2 - Provides API for querying MaxMind GeoIP database.¹⁰
 - ▶ multiprocessing - Process-based parallelism for faster execution.¹¹

Dataset Composition

- ▶ Tranco dataset (ID: 4K5KX) provided initial collection of popular domains.
 - ▶ Contains over 4.6 million domains ranked by popularity.
 - ▶ This dataset is based on network activity between October 11th and November 9th.
- ▶ socket module in Python performed DNS lookup to collect IPv4 addresses.
 - ▶ IPv4 addresses were required for querying MaxMind GeoIP database.
 - ▶ Multiple IP addresses are returned from this request; primary is used for GeoIP.
- ▶ MaxMind's GeoIP dataset estimates the physical location of servers.
 - ▶ The database was downloaded and queried using the Python geoip2 module.
 - ▶ The entire return was analyzed; however, the country name was the primary info.
 - ▶ This is the cornerstone of comparing China's servers from other country's servers.

Security Measurements

- ▶ At this stage, the multiprocessing module in Python was leveraged to process multiple requests simultaneously in parallel for larger sample sizes.
- ▶ Python requests module identified if HTTPS is available for each domain.
 - ▶ A request was sent to each domain for an HTTPS version of the site.
 - ▶ The compatibility of a domain with HTTPS was captured if HTTPS message was returned instead of simply HTTP.
- ▶ ssl, OpenSSL, and socket were used in conjunction to acquire TLS/SSL certs.
 - ▶ The certificate validity was confirmed to ensure the site appropriately used HTTPS.
 - ▶ The contents of the certificate were further used for analysis.
- ▶ ssl and socket together identified the version of TLS being used (1.2 or 1.3).

Results

Composition of Final Dataset

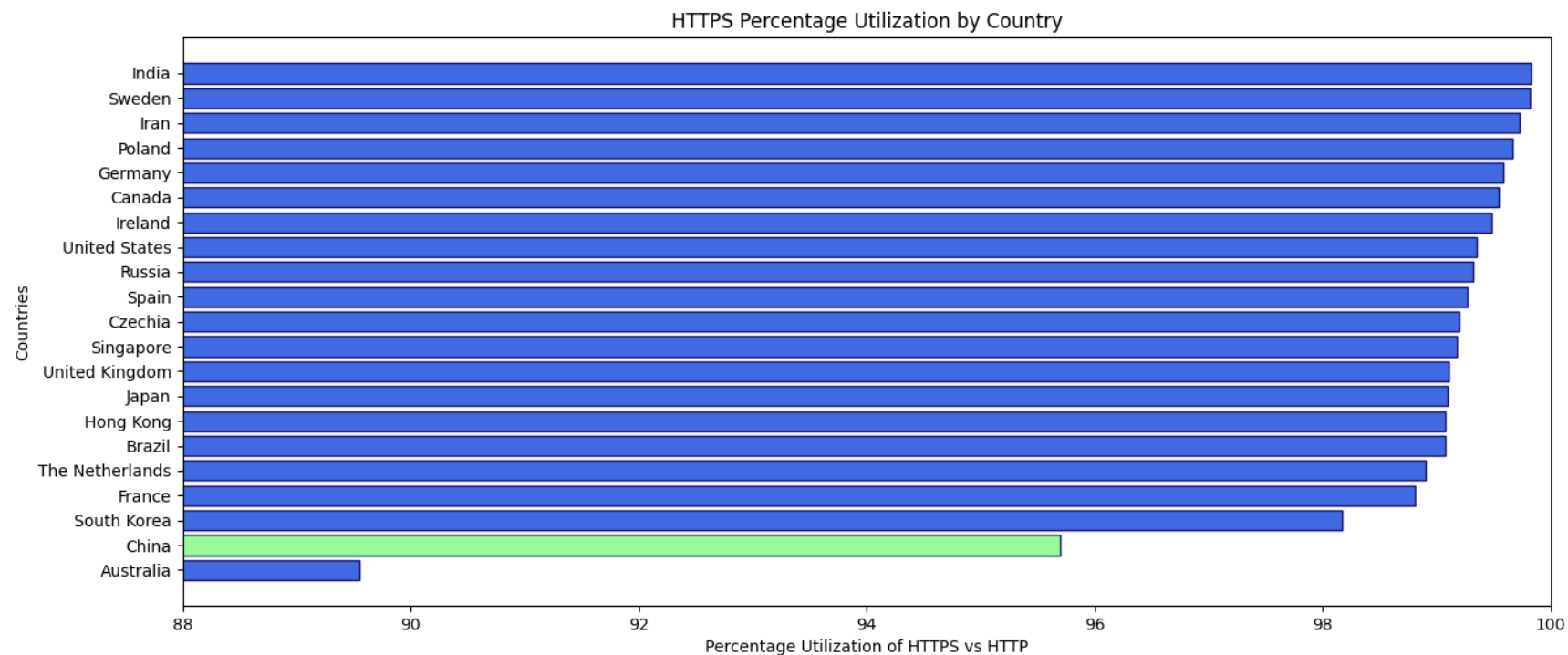
- ▶ The dataset was initially populated with the most popular 4.6+ million domains over the months of October-November from the Tranco database.
- ▶ After each measurement step, only domains which succeeded the previous step were evaluated within the next.
- ▶ These are the number of domains which...
 - ▶ successfully passed the Tranco import stage: 4,644,227
 - ▶ successfully returned an IPv4 address: 271,614
 - ▶ successfully were geolocated: 229,717
 - ▶ successfully confirmed HTTPS compatibility: 111,318
 - ▶ successfully passed the TLS version check: 35,144
 - ▶ successfully provided valid TLS/SSL certificates: 23,545

These steps created a bottle-neck in the size of the dataset due to the speed at which domains could be processed

Rank	Country	Initial Size	Rank	Country	Initial Size
1	United States	119,120	7	Canada	5,597
2	Germany	13,382	8	Japan	5,408
3	Russia	12,982	9	United Kingdom	5,229
4	The Netherlands	7,378	10	Ireland	3,155
5	France	6,993	11	India	2,960
6	China	6,400	...	...	...

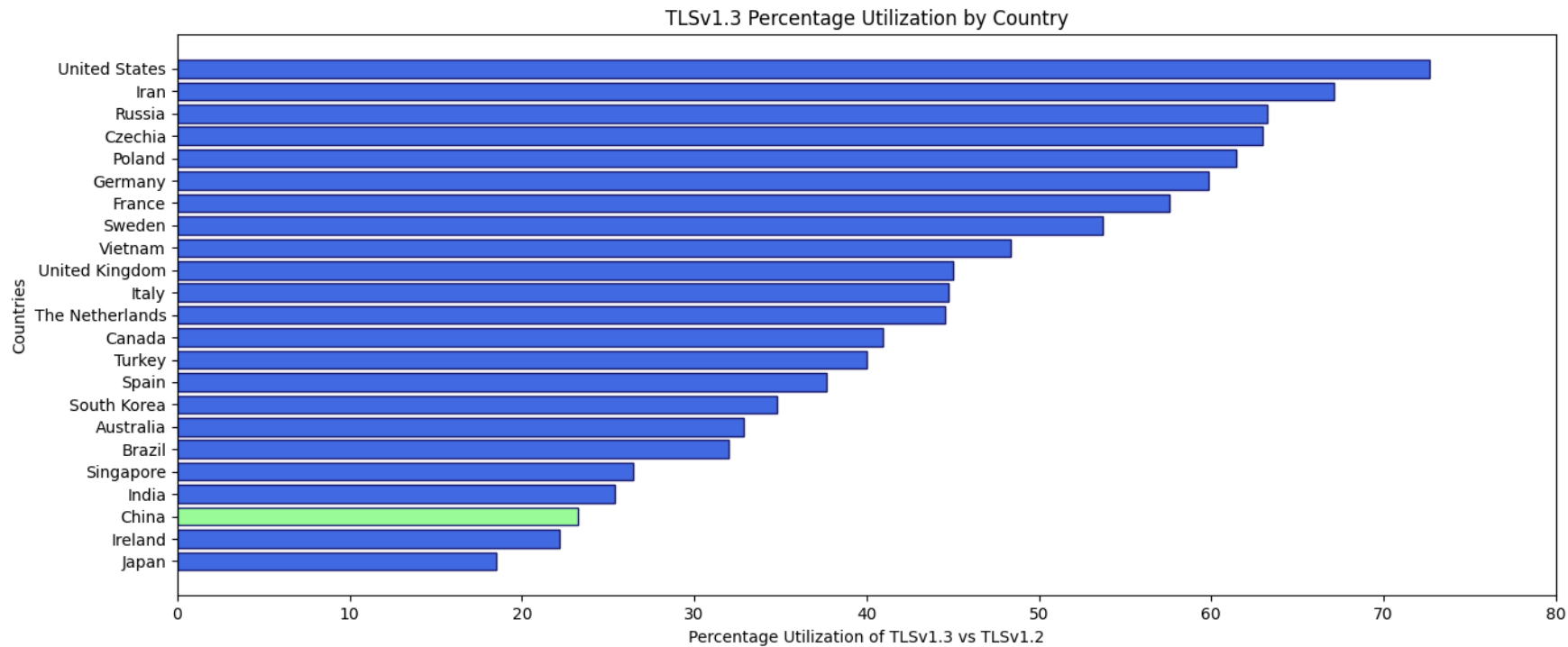
Sample-Size by Country

- These are the sample sizes after the GeoIP measurement successfully returned.
- A total of 203 countries were represented within this dataset, and only the most popular 11 are shown above.



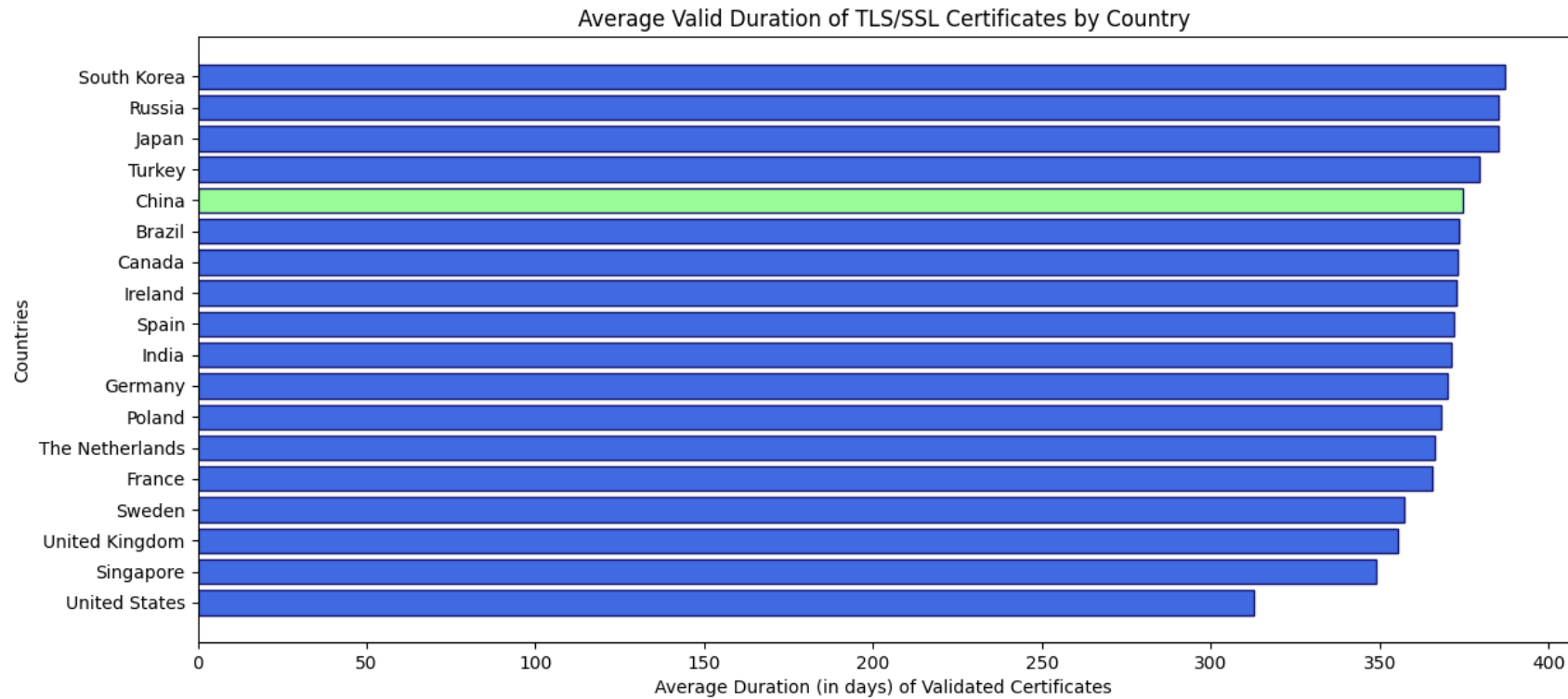
Proportion of Websites Utilizing HTTPS by Country

- Domains located in China rank relatively low in HTTPS Percentage Utilization at 95.7%.
- The only countries shown above were those represented by at least 500 unique domains.



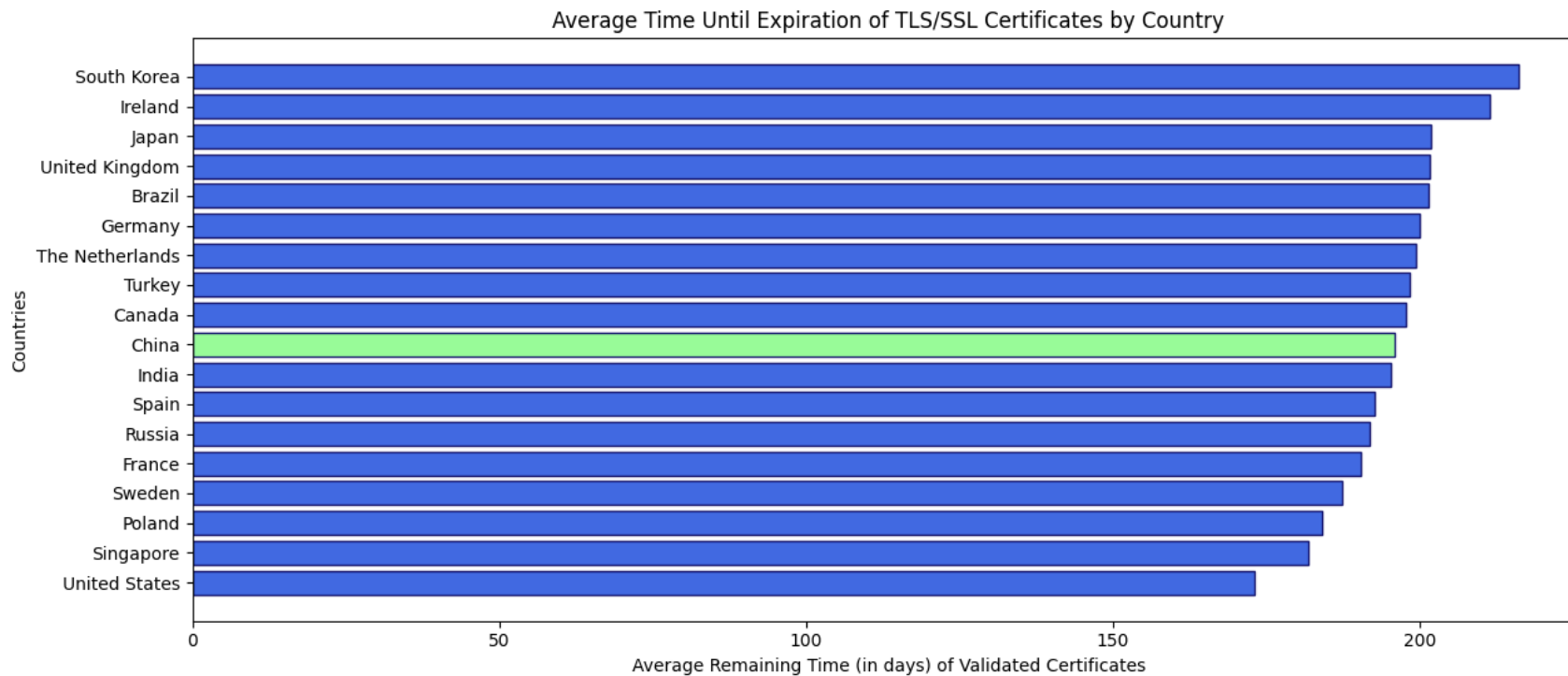
TLS Version for HTTPS Websites by Country

- Domains located in China also rank relatively low in utilization of TLSv1.3 at 23.2%.
- The only countries shown above were those represented by at least 100 unique domains.



Total Duration of Valid TLS/SSL Certificates by Country

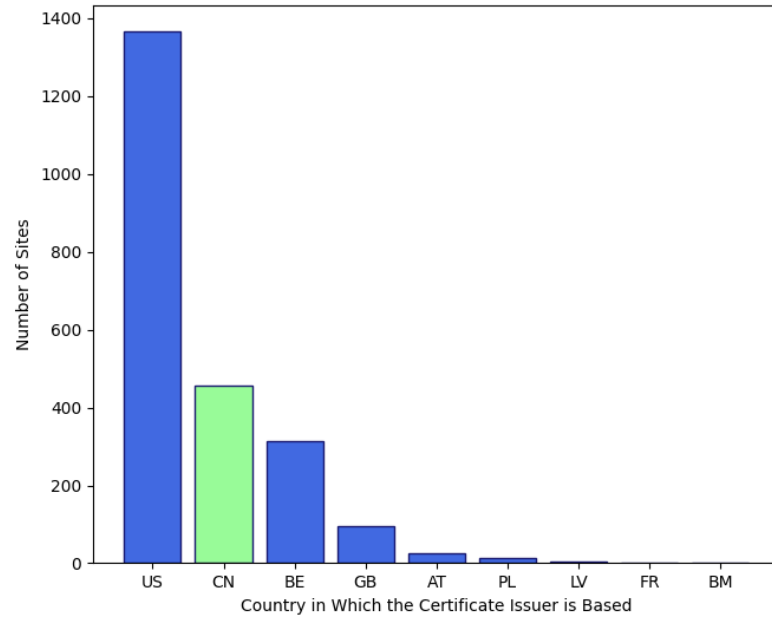
- There does not appear to be significant spread in certificate duration with all domains having around 1-year durations.
- The United States might be the only outlier with an average duration more than 1 month away from the 1-year mean.



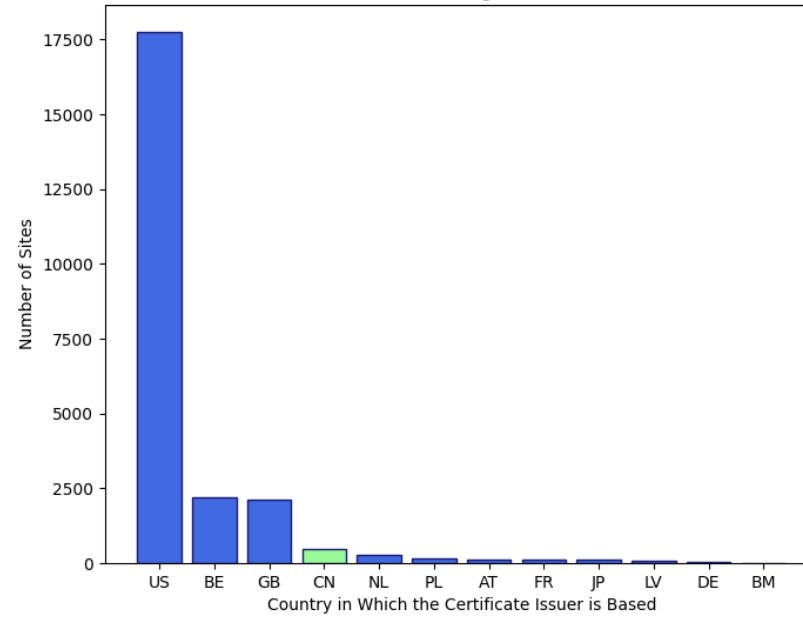
TLS/SSL Certificate Expiration Dates by Country

- There does not appear to be significant spread in remaining certificate duration.
- The only countries shown in the last two charts were those represented by at least 100 unique domains.

China Websites: Number of Sites Utilizing Certificate Issuers Based in Each Country



All Countries Websites: Number of Sites Utilizing Certificate Issuers Based in Each Country



Utilized Certificate Issuers

- United States Certificate Issuers are the most utilized issuers, even in China.
- However, Chinese Certificate Issuers are more utilized within China than on average worldwide.

Discussion

Some Reasons for Disparity in HTTPS Adoption Statistics

- ▶ As explored within the “Timeline of HTTPS Adoption,” there exist historical reasons contributing to the disparity in HTTPS adoption in China.¹²
 - ▶ Restricted access to popular HTTPS websites such as Wikipedia and BBC lowers the number of HTTPS services available within China.
 - ▶ Very popular, high traffic sites such as QQ and Baidu (which did not support HTTPS at some points in time) represent a significant portion of the unencrypted traffic.
- ▶ China’s country-wide firewall is sometimes incompatible with TLS encryption.¹³
 - ▶ The firewall enables government regulation of illicit activity online such as violence and gambling. It also bolsters international cyber and network security.
 - ▶ Some firewall filtering methods depend on the ability to access the unencrypted content of a network message.

Security Risks Of Low HTTPS Adoption Rate

- ▶ Measurements such as those captured within this study are utilized to identify potential vulnerabilities and security risks in real-world networks.
- ▶ For example, Security Analysis of the Chinese Web: How well is it protected? analyzes the implementation of insecure SSL/TLS implementations (among other web characteristics).¹⁴
 - ▶ Web-crawling was performed to acquire a representative sample of the Chinese web within a dataset of 10,000 websites. This study comparatively utilized a set of 6,400 candidate websites hosted within China.
 - ▶ Security policies were analyzed and measured, and these results were compared to websites hosted outside of China to identify unique security risks.

Internet Security in China is Improving

- ▶ New Chinese Cybersecurity and Data Privacy Requirements describes the Cybersecurity Law of the People's Republic of China.¹⁵
 - ▶ Taking effect in June 2017, this set of laws and regulations establishes national standards for network security in China.
 - ▶ Encryption products in particular are receiving approval to aid in public interest, the national economy, and people's livelihoods as of January 2020.

Potential Next Steps and Future Work

- ▶ This study can be expanded by experimenting with many additional pursuits:
 - ▶ Perform similar analysis for nearby countries such as Japan or South Korea.
 - ▶ Explore HTTPS utilization differences at a finer scale such as comparing cities instead of comparing countries (e.g. Shenzhen vs Beijing). Explore sub-domains or paths that may implement HTTP/HTTPS differently.
 - ▶ Explore tangential security topics such as utilization of tunnelling or cookies.
 - ▶ Compare websites by industry or content instead of by geography.
 - ▶ Examine the TLS handshake itself using tools such as scapy instead of simply analyzing the end-result of the TLS connection.¹⁶
 - ▶ Explore artifacts contained within the pages such as links, images, and frames to other domain's content.

References and Resources

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Thank you!

- More detailed information relevant to this study will be contained within the final project paper write-up.

References and Resources

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[15] "New Chinese Cybersecurity and Data Privacy Requirements," Jones Day Insights.

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